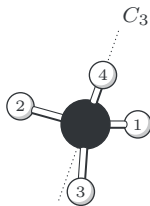


rigid methane, $\mathbf{X}_0$



the three C_2 axes

$H = T_d(\mathbf{M}) \cong S_4$: each relabelling of 1–4 keeps the orbit even ones as rotations: $H_{\text{rot}} = T \cong A_4$ (8 C_3 , 3 C_2)
 odd ones with the star: parity, this paper's addition
 spin $16 = 5 A_1 \oplus E \oplus 3 T_2$ under H
 $= 5 A \oplus {}^1E \oplus {}^2E \oplus 3 T$ under T

this paper: species, parity, states

$\mathcal{H}_{\text{spin}}$: one line per state

Albert et al., Table II

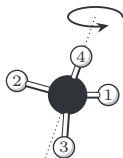
		5 A_1 , under T : 5 A		
A_1^-, A_2^+	5 each			(A, A, 1): 5
		E: ${}^1E \oplus {}^2E$		
E^-, E^+	1 each			$({}^1E, {}^2E, 1)$: 1
				$({}^2E, {}^1E, 1)$: 1
		3 T_2 : 3 T		
T_2^-, T_1^+	3 each			(T, T, 3): 3



the $T \otimes T$ invariant: Schmidt rank 3

a third of a turn about the C–H₄ axis: a loop in $SO(3)/T$, lifted it ends at rg

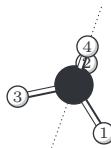
the fibre after the loop



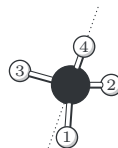
0



$2\pi/9$



$4\pi/9$



$2\pi/3$

same position, 1, 2, 3 cycled

